

An Exact Locked-Permutation Certificate for Property B: $m(5) \geq 33$

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Abstract

Let $m(n)$ be the minimum number of edges in an n -uniform hypergraph that is not two-colourable. The previously established bounds at uniformity five are $32 \leq m(5) \leq 51$. We prove that every 5-uniform hypergraph with at most 32 edges is two-colourable, and hence

$$33 \leq m(5) \leq 51.$$

The proof refines the random greedy method by locking a small, deterministically selected set of low-degree vertices into prescribed positions of a random permutation. Pair compression, balanced-colouring counting, and an unconditioned greedy estimate reduce a hypothetical obstruction to five possible vertex counts, 21 through 25. Degree and codegree selection lemmas then reduce those cases to finite trace-profile calculations on at most five selected vertices. The exact computations include 3,081 certificate records for vertex counts 21 and 22, together with six labelled clique partitions of K_4 and 32 of K_5 . A separately written implementation regenerates the finite universes and re-derives the probability coefficients. We also determine a precise ceiling for the current certificate families: exhaustive optimization of their locked positions does not extend this argument to $m(5) \geq 34$.

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1 Introduction

An n -uniform hypergraph has *Property B* if its vertices admit a red–blue colouring with no monochromatic edge. For integers n and v , write

$$m(n, v) := \min\{|E(H)| : H \text{ is an } n\text{-uniform non-two-colourable hypergraph on } v \text{ vertices}\},$$

with $m(n, v) = \infty$ if no such hypergraph exists, and put $m(n) = \min_v m(n, v)$. Equivalently, Property B is the feasibility version of monotone NAE-SAT or set splitting with every constraint of size n .

The exact values of $m(n)$ are known only for $n \leq 4$ [7]; in particular, $m(4) = 23$ was established by an exhaustive computation of Östergård [8]. At uniformity five, Abbott and Hanson constructed a 51-edge obstruction [1]. On the lower side, Aglave, Amarnath, Shannigrahi and Singh proved

$m(5) \geq 29$ [2], and Grill and Linzmayer subsequently proved $m(5) \geq 32$ [6]. Thus the interval immediately preceding this work was

$$32 \leq m(5) \leq 51.$$

The 2026 overview of Grill and Linzmayer cites that 2024 lower-bound preprint [7]. To the best of our knowledge, as of August 23, 2026 no publicly available proof of $m(5) \geq 33$ had appeared. A dated primary-source search is included in the supplementary artifact; as usual, such a search cannot rule out unpublished work.

Our main result improves the lower endpoint by one.

Theorem 1. *Every 5-uniform hypergraph with at most 32 edges has Property B. Consequently,*

$$m(5) \geq 33.$$

The proof belongs to the random greedy line initiated by Pluhár [9], developed asymptotically by Cherkashin and Kozik [4], and adapted to exact small parameters by Grill and Linzmayer [6]. The comparison with the last of these is precise. Their unconditioned discrete estimate and low-degree conditioning prove two-colourability for hypergraphs with at most 31 edges. Here we give a self-contained normalization for 32 edges, derive a general formula for locking any fixed number of vertices, and combine three-, four- and five-lock certificates with structural selection arguments. The earlier $m(5) \geq 32$ computation is not used as a premise.

The computer-assisted portion of the proof of Theorem 1 is deliberately smaller than an enumeration of hypergraphs. It performs the following tasks only:

1. enumerate a safe superset of the integer trace profiles of three selected vertices;
2. enumerate clique partitions of K_4 and K_5 that encode traces on pairwise-codegree-one selected sets;
3. evaluate a displayed rational expression at finite, explicitly listed position assignments; and
4. compare each reduced fraction with one.

No SAT solver, mixed-integer solver, stochastic search, or floating-point acceptance test enters Theorem 1. Proposition 6 states the complete machine-checked interface; Sections 5 and 7 explain how it is independently replayed.

2 A pair-covered 32-edge normal form

We work with finite simple hypergraphs. Duplicate edges may always be discarded. The following standard pair-identification reduction appears in the exact Property-B literature, including Goldberg and Russell [5]; we include the proof because the repeated padding step makes the present theorem independent of prior numerical bounds.

Lemma 2 (pair compression and padding). *If a non-two-colourable 5-uniform hypergraph with at most 32 edges exists, then there is a non-two-colourable simple 5-uniform hypergraph with exactly 32 edges in which every pair of vertices occurs in an edge.*

Proof. A 5-uniform hypergraph on at most eight vertices is two-colourable: split its vertex set into two classes of size at most four. Hence an obstruction has at least nine vertices. Since $\binom{9}{5} = 126 > 32$, distinct edges can be added until there are exactly 32; adding constraints preserves non-two-colourability.

Suppose that vertices x, y occur together in no edge. Identify x and y . No image edge loses a vertex, so every image edge still has size five. Any two-colouring of the quotient pulls back by giving x and y the quotient colour, so the quotient remains non-two-colourable. After coincident image edges are discarded it has at most 32 edges. If its order is at most eight, the preceding split is a contradiction; otherwise it has at least nine vertices and can again be padded to 32 distinct edges. Repeating this operation strictly decreases the order at every identification, so it terminates at the claimed pair-covered hypergraph. $\square$

Let H henceforth be such a normal-form hypergraph, and write

$$v = |V(H)|, \quad d(x) = |\{e : x \in e\}|, \quad \lambda_{xy} = |\{e : \{x, y\} \subseteq e\}|.$$

Pair coverage gives

$$\lambda_{xy} \geq 1, \quad \binom{v}{2} \leq 32 \binom{5}{2} = 320, \quad (1)$$

so $v \leq 25$.

A uniformly chosen colouring with colour-class sizes $\lfloor v/2 \rfloor$ and $\lceil v/2 \rceil$ makes any fixed 5-edge monochromatic with probability

$$\frac{\binom{\lfloor v/2 \rfloor}{5} + \binom{\lceil v/2 \rceil}{5}}{\binom{v}{5}}.$$

The first-moment method therefore gives the familiar bound

$$m(5, v) \geq \frac{\binom{v}{5}}{\binom{\lfloor v/2 \rfloor}{5} + \binom{\lceil v/2 \rceil}{5}}; \quad (2)$$

see also [5, 6]. For $10 \leq v \leq 18$, the ceilings of the right-hand side are

$$126, 66, 66, 48, 48, 39, 39, 34, 34,$$

respectively; orders at most nine are immediate from the same balanced split. Hence

$$19 \leq v \leq 25. \quad (3)$$

We shall repeatedly use

$$d(x) \geq \left\lceil \frac{v-1}{4} \right\rceil, \quad \sum_x d(x) = 5|E(H)| = 160, \quad (4)$$

and the exact local pair excess

$$\rho_x := \sum_{y \neq x} (\lambda_{xy} - 1) = 4d(x) - (v-1). \quad (5)$$

The first inequality in (4) follows because every edge through x covers four of the $v-1$ pairs incident with x .

3 The locked-permutation bound

Choose an ordered set $S = (s_1, \dots, s_s)$ of *locked* vertices, where $0 \leq s \leq 5$ and $f := v - s \geq 9$, and distinct positions $p_1, \dots, p_s \in [v]$. Assign the other vertices uniformly to the free positions. For $A \subseteq [s]$, let

$$l_A = |\{e \in E(H) : e \cap S = \{s_i : i \in A\}\}|.$$

For a position k , let $b(k)$ and $a(k)$ be the numbers of free positions before and after k . Write $A \prec k$ if every lock in A precedes k , and $A \succ k$ if every lock in A follows k . Empty conditions are vacuous, and binomial coefficients outside their natural range are zero.

Run Pluhár's greedy algorithm: begin with every vertex red, process the permutation from first to last, and turn the current vertex blue precisely when it is last in an otherwise-red edge. No red edge survives. If a blue edge survives, its earliest vertex is simultaneously last in a red witness edge and first in the blue edge. These two edges meet only at that vertex; call it *critical*.

For a free position k , union bounds for being last, first, or both are

$$L_k = \sum_{\substack{A \prec k \\ |A| \leq 4}} l_A \frac{5 - |A|}{f} \frac{\binom{b(k)}{4 - |A|}}{\binom{f-1}{4 - |A|}}, \quad (6)$$

$$F_k = \sum_{\substack{A \succ k \\ |A| \leq 4}} l_A \frac{5 - |A|}{f} \frac{\binom{a(k)}{4 - |A|}}{\binom{f-1}{4 - |A|}}, \quad (7)$$

$$B_k = \sum_{\substack{A \cap B = \emptyset \\ A \prec k, B \succ k \\ |A|, |B| \leq 4}} (l_A l_B - \mathbf{1}_{A=B} l_A) \frac{\binom{b(k)}{4 - |A|} \binom{a(k)}{4 - |B|}}{f \binom{f-1}{4 - |A|} \binom{f-1 - (4 - |A|)}{4 - |B|}}. \quad (8)$$

Trace classes with $|A| = 5$ contain no free vertex and therefore do not contribute at a free position. For a locked position $k = p_i$, the corresponding expressions are

$$L_k = \sum_{\substack{i \in A \\ A \setminus \{i\} \prec k}} l_A \frac{\binom{b(k)}{5 - |A|}}{\binom{f}{5 - |A|}}, \quad (9)$$

$$F_k = \sum_{\substack{i \in A \\ A \setminus \{i\} \succ k}} l_A \frac{\binom{a(k)}{5 - |A|}}{\binom{f}{5 - |A|}}, \quad (10)$$

$$B_k = \sum_{\substack{A \cap B = \{i\} \\ A \setminus \{i\} \prec k, B \setminus \{i\} \succ k}} (l_A l_B - \mathbf{1}_{A=B} l_A) \frac{\binom{b(k)}{5 - |A|} \binom{a(k)}{5 - |B|}}{\binom{f}{5 - |A|} \binom{f - (5 - |A|)}{5 - |B|}}. \quad (11)$$

Lemma 3 (locked-permutation lemma). *With the notation above,*

$$\Pr(\text{greedy failure}) \leq \mathcal{Q}(H, S, p) := \sum_{k=1}^v \min\{L_k, F_k, B_k\}. \quad (12)$$

In particular, $\mathcal{Q}(H, S, p) < 1$ implies that H has Property B.

Proof. For an edge of type A at a free position, the factor $(5 - |A|)/f$ chooses its occupant and the binomial ratio places its remaining free vertices on the required side; this proves (6) and (7). A compatible ordered witness pair must have disjoint locked traces. After its common free vertex occupies k , the other free vertices of the two edges occupy disjoint earlier and later slots, giving (8). The category product is an upper bound because it may also count edge pairs having an additional common free vertex. The diagonal subtraction only forbids using one edge twice.

At a locked position the common vertex is fixed, yielding (9)–(11); now the two locked traces must meet exactly in the locked vertex. Criticality at k is contained in each of the last-edge, first-edge and paired events, so its probability is at most their minimum. Summing over positions proves (12). $\square$

The locked set and its position assignment may depend deterministically on the fixed hypergraph and its trace profile. Only after this choice do we randomize the remaining vertices. Consequently, different profiles may use different assignments without any additional union bound.

The zero-lock specialization of Lemma 3 is useful at the two smallest remaining orders. Let γ be the number of *ordered* pairs of distinct edges meeting in exactly one vertex. Then

$$U_0(v, \gamma) = \sum_{k=1}^v \frac{1}{v} \min \left\{ 160 \frac{\binom{k-1}{4}}{\binom{v-1}{4}}, 160 \frac{\binom{v-k}{4}}{\binom{v-1}{4}}, \gamma \frac{\binom{k-1}{4} \binom{v-k}{4}}{\binom{v-1}{4} \binom{v-5}{4}} \right\}. \quad (13)$$

Using $\gamma \leq 32 \cdot 31 = 992$ gives

$$U_0(19, 992) = \frac{222656}{230945} < 1.$$

For $v = 20$, the pair-incidence surplus is

$$R = 320 - \binom{20}{2} = 130.$$

Moreover,

$$\sum_{x < y} \binom{\lambda_{xy}}{2} = \sum_{\{e, f\}} \binom{|e \cap f|}{2} \geq R.$$

Distinct 5-edges meet in at most four vertices. Thus at least $\lceil 130/6 \rceil = 22$ unordered edge pairs meet in at least two vertices, so $\gamma \leq 992 - 2 \cdot 22 = 948$, and

$$U_0(20, 948) = \frac{20874}{20995} < 1.$$

It remains to exclude

$$21 \leq v \leq 25. \quad (14)$$

4 Finite trace profiles and certification

This section specifies the finite computation in enough detail for an independent implementation.

4.1 Three-lock profiles

For three selected vertices, counts are stored in binary-mask order

$$(l_{000}, l_{001}, l_{010}, l_{011}, l_{100}, l_{101}, l_{110}, l_{111}), \quad (15)$$

where the i th selected vertex is represented by bit 2^{i-1} . Thus, for example, l_{011} counts edges containing the first two selected vertices and not the third.

Lemma 4 (complete safe profile universe). *Fix selected degrees (d_1, d_2, d_3) . Every trace profile of a pair-covered 32-edge hypergraph is produced by the following enumeration: choose nonnegative integers*

$$t = l_{111}, \quad q_{12} = l_{011}, \quad q_{13} = l_{101}, \quad q_{23} = l_{110},$$

derive

$$\begin{aligned} l_{001} &= d_1 - q_{12} - q_{13} - t, \\ l_{010} &= d_2 - q_{12} - q_{23} - t, \\ l_{100} &= d_3 - q_{13} - q_{23} - t, \\ l_{000} &= 32 - \sum_{\alpha \neq 000} l_{\alpha}, \end{aligned}$$

and retain the tuple precisely when every entry is nonnegative and

$$q_{12} + t \geq 1, \quad q_{13} + t \geq 1, \quad q_{23} + t \geq 1.$$

The output may contain unrealizable profiles, but cannot omit a realizable one.

Proof. For an actual profile, $t, q_{12}, q_{13}, q_{23}$ are among the loop choices. The three displayed degree equations uniquely force the singleton counts, the 32-edge total uniquely forces l_{000} , and pair coverage gives the three final inequalities. Hence every actual profile is retained. Since no constraints involving unselected vertices are imposed, the output is in general a safe superset rather than a realizability characterization. $\square$

If $d_1 \leq d_2 \leq d_3$ are the three smallest degrees, then

$$\begin{aligned} d_1 &\geq \left\lceil \frac{v-1}{4} \right\rceil, \\ d_1 + d_2 + (v-2)d_3 &\leq 160. \end{aligned} \tag{16}$$

Looping over all ordered integer triples satisfying (16) therefore gives the complete degree universe used below.

4.2 Clique partitions

Lemma 5 (trace partitions). *If every pair in a selected set S has codegree one, then the traces $e \cap S$ of size at least two partition $E(K_S)$ into edge sets of cliques. Once this clique partition and the selected degrees are fixed, all singleton and empty trace counts are forced.*

Proof. A trace T contributes precisely the pairs in $\binom{T}{2}$. Every selected pair occurs in one trace, so these clique edge sets are disjoint and cover $E(K_S)$. Subtracting the incidences of the non-singleton traces from each selected degree determines the singleton counts, and the total edge count then determines the empty count. $\square$

The clique-partition generator is complete by construction: choose the lexicographically first uncovered pair, branch over every clique containing that pair whose edge set is disjoint from the already covered pairs, and recurse. It returns six labelled partitions of K_4 and 32 labelled partitions of K_5 . The K_4 partitions have the three types

$$K_4, \quad K_3 + 3K_2, \quad 6K_2.$$

4.3 Pseudocode and position menus

The primary computation is summarized below. All arithmetic in LOCKEDBOUND is integer arithmetic and reduced rational arithmetic using equations (6)–(12).

```

PROFILE3(d1,d2,d3):
  for t,q12,q13,q23 in their finite degree-bounded ranges:
    derive the three singleton counts from d1,d2,d3
    derive the empty count from the 32-edge total
    if all counts are nonnegative and each selected pair is covered:
      emit the eight-count tuple

CERTIFY(v, profile, menu):
  best := +infinity
  for injective labelled position assignment p in menu:
    best := min(best, LOCKEDBOUND(v,profile,p))
  return best and the lexicographically first minimizing p

CLIQUEPARTITIONS(s, covered):
  if every pair of K_s is covered: emit the selected cliques
  otherwise let xy be the first uncovered pair
  for each clique C containing xy and no covered pair:
    CLIQUEPARTITIONS(s, covered union E(C))

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For a tuple $(a_1, \dots, a_s)$, “all permutations” below means all label-to-position bijections onto that position set. The complete menus are

order	profile family	allowed assignments
21	three lowest	all permutations of $(10, 11, 12), (1, 2, 3), (11, 12, 13), (1, 2, 12)$
22	three lowest/Ramsey	all permutations of $(10, 11, 12), (1, 2, 3), (1, 2, 11), (1, 2, 13), (1, 11, 13)$
23	three lowest	all permutations of $(11, 12, 13)$
23	degrees $(6, 6, 7, 7)$	all permutations of $(10, 11, 12, 13)$ or $(12, 13, 14, 15)$
24	degrees $(6, 6, 6, 6)$	all permutations of $(11, 12, 13, 14)$
25	degrees $(6, 6, 6, 6, 6)$	all permutations of $(11, 12, 13, 14, 15)$

For the five-lock $v = 23$ family with degrees $(6, 7, 7, 7, 7)$, the complete menu is the following list of eleven *ordered* assignments:

$$\begin{aligned}
& (13, 16, 17, 18, 14), (11, 4, 3, 2, 1), (16, 15, 17, 18, 14), \\
& (10, 4, 3, 1, 2), (15, 16, 17, 18, 14), (10, 4, 3, 2, 1), \\
& (12, 15, 16, 17, 14), (17, 15, 16, 18, 14), \\
& (12, 16, 17, 18, 14), (10, 15, 16, 17, 14), (12, 4, 3, 2, 1).
\end{aligned} \tag{17}$$

No unlisted position is used in the proof. Among the six $v = 21$ position bases screened during optimization, an exhaustive exact check of all base subsets shows that no three-base subset attains the worst-case value 9599/9724; the four-base menu displayed above does. This is a minimality statement within that candidate family, not among all conceivable three-lock position sets.

Proposition 6 (exact finite certificate). *The algorithms above, evaluated with exact rational arithmetic, give the following exhaustive results.*

v	family	profiles	residuals	largest closing bound
21	three lowest	1,864	0	9599/9724
22	three lowest	1,000	3	—
22	Ramsey independent	2	0	91231/92378
22	Ramsey triangle	215	0	456463/461890
23	degrees (6, 6, 6)	192	0	245/247
23	degrees (6, 6, 7)	220	2	—
23	degrees (6, 7, 7)	263	17	—
23	admissible K_4 partitions	3	0	8315/8398
23	all K_5 partitions	32	0	45784/51051
24	all K_4 partitions	6	0	16671/16796
25	favourable K_5 partitions	7	0	4188/4199

A dash denotes a family containing residual profiles, for which no family-wide strict closing bound exists. Here “residual” means that no assignment in the stated menu gives a value strictly below one. In the three $v = 22$ residuals the selected degrees are (7, 7, 7), the pair-codegree multiset is (1, 1, 2), and $l_{111} = 0$. In the $v = 23$ residuals, the two (6, 6, 7) profiles have all selected codegrees one, and all 17 (6, 7, 7) profiles have selected codegrees at most two.

The proposition is a direct output assertion of the primary generator and is independently recomputed by a second implementation. To make the scale and encoding concrete, three representative closing rows are

family	degrees	counts in (15); positions	bound
$v = 21$ worst	(7, 7, 7)	(15, 5, 4, 1, 4, 1, 2, 0); (1, 2, 12)	9599/9724
$v = 22$ independent	(7, 7, 7)	(14, 5, 5, 1, 5, 1, 1, 0); (1, 2, 3)	91231/92378
$v = 22$ triangle	(7, 7, 7)	(17, 3, 3, 2, 3, 2, 2, 0); (1, 11, 13)	456463/461890

The complete records, not merely these representatives, are included in the artifact described in Section 7.

5 The five remaining orders

We now combine Proposition 6 with deterministic selection arguments. This separation is important: the computer checks profiles and fractions, while the proofs below show that every hypothetical hypergraph contains a selected set belonging to one of the checked families.

5.1 Order 21

Let $d_1 \leq d_2 \leq d_3$ be the three smallest degrees. From (16),

$$d_1 \geq 5, \quad d_1 + d_2 + 19d_3 \leq 160.$$

The complete list is

$$(5, 5, 5), (5, 5, 6), (5, 5, 7), (5, 6, 6), (5, 6, 7), \\ (5, 7, 7), (6, 6, 6), (6, 6, 7), (6, 7, 7), (7, 7, 7).$$

Lemma 4 gives 1,864 profiles, all closed by the first line of Proposition 6. Therefore $v = 21$ is impossible.

5.2 Order 22

The complete lowest-degree triple list is

$$(6, 6, 6), (6, 6, 7), (6, 7, 7), (7, 7, 7),$$

which produces 1,000 safe profiles. The menu closes 997. The three residuals are exactly the labelings

$$\begin{aligned} (15, 4, 4, 2, 5, 1, 1, 0), \\ (15, 4, 5, 1, 4, 2, 1, 0), \\ (15, 5, 4, 1, 4, 1, 2, 0). \end{aligned} \tag{18}$$

Their best menu value is $1395847/1385670 > 1$; they are not claimed to be realisable. What matters is the common structural consequence: the selected vertices have degrees $(7, 7, 7)$, pair-codegrees $(1, 1, 2)$ in some order, and no edge contains all three.

Because the selected vertices were the three smallest, a residual obstruction has minimum degree seven. The total excess above seven is

$$160 - 22 \cdot 7 = 6,$$

so at most six vertices have degree greater than seven and at least 16 have degree exactly seven. On the degree-seven vertices define

$$xy \in E(G) \iff \lambda_{xy} \geq 2.$$

Because the minimum degree is seven, any three degree-seven vertices may be chosen as the three lowest-degree vertices in the preceding enumeration. Every graph on at least six vertices contains either an independent triple or a triangle, since $R(3, 3) = 6$. In the independent case, pair coverage forces all three selected codegrees to equal one; in the triangle case all three are at least two. Lemma 4 produces exactly two safe profiles in the first class and 215 in the second, and Proposition 6 closes both classes strictly. Hence $v = 22$ is impossible.

5.3 Order 23

The degree constraints leave only

$$(d_1, d_2, d_3) = (6, 6, 6), (6, 6, 7), (6, 7, 7).$$

The central three-lock calculation has profile/residual counts 192/0, 220/2 and 263/17, respectively. Thus a residual obstruction has no three degree-six vertices. It must nevertheless have a degree-six vertex, since $23 \cdot 7 = 161 > 160$. If there are a degree-six vertices, then

$$\sum_x (d(x) - 7) = -1.$$

With $1 \leq a \leq 2$, the only possible degree sequences are therefore

$$(6^2, 7^{20}, 8), \quad (6, 7^{22}). \tag{19}$$

Ties among degree-seven vertices may be broken arbitrarily, so the residual conclusions apply to every degree-seven vertex or pair chosen below.

First consider $(6^2, 7^{20}, 8)$. Let u, v be the degree-six vertices. For every degree-seven vertex y , the selected triple (u, v, y) must be one of the two $(6, 6, 7)$ residual profiles; otherwise it is already closed. Proposition 6 therefore gives

$$\lambda_{uv} = \lambda_{uy} = \lambda_{vy} = 1 \quad \text{for every such } y.$$

Let e be the unique edge containing u, v . At most three of the 20 degree-seven vertices lie in e , so at least 17 lie outside it. Among those 17, some pair y, z has $\lambda_{yz} = 1$: if every such pair were repeated, then each vertex in this set would have local excess at least 16, contradicting

$$\rho_y = 4 \cdot 7 - 22 = 6.$$

Thus $S = \{u, v, y, z\}$ is pairwise codegree one, and the unique trace containing u, v is exactly $\{u, v\}$ because $y, z \notin e$. Of the six labelled K_4 clique partitions, precisely three have this property. The four-lock row of Proposition 6 closes all three. The $6K_2$ row uses positions $(12, 13, 14, 15)$ and has value $8315/8398$; the two $K_3 + 3K_2$ rows have value $4152/4199$. All are strict.

Now consider $(6, 7^{22})$, and let u be the degree-six vertex. Applying the $(6, 7, 7)$ residual consequence to u and every pair of degree-seven vertices shows that every pair-codegree in H is at most two. Pair coverage therefore makes every codegree one or two. Define the repeated-pair graph

$$xy \in E(G) \iff \lambda_{xy} = 2.$$

Equation (5) now equals the graph degree, so G has degree sequence $(2, 6^{22})$ and 67 edges. Let $A = N_G(u)$ and $B = V(H) \setminus (\{u\} \cup A)$. Then $|A| = 2$, $|B| = 20$, and for $R = \{u\} \cup A$ we have $e(G[R]) \in \{2, 3\}$. The number of graph edges incident with R is

$$\sum_{x \in R} d_G(x) - e(G[R]) = 14 - e(G[R]),$$

whence

$$e(G[B]) = 67 - (14 - e(G[R])) \in \{55, 56\}.$$

The Caro–Wei inequality [3, 10], followed by Cauchy–Schwarz, gives

$$\alpha(G[B]) \geq \sum_{x \in B} \frac{1}{d_B(x) + 1} \geq \frac{|B|^2}{2e(G[B]) + |B|} \geq \frac{400}{132} = \frac{100}{33} > 3.$$

Choose four independent vertices in B . Together with u they form a five-set with every pair-codegree one. Lemma 5 gives one of the 32 labelled K_5 partitions, and the explicit menu (17) closes all 32 with maximum $45784/51051 < 1$. This excludes both sequences in (19).

5.4 Order 24

Every degree is at least six, and the total excess above six is

$$160 - 24 \cdot 6 = 16.$$

Hence at least eight vertices have degree six; call their set D . For $x \in D$, equation (5) gives $\rho_x = 1$. The graph induced on D by repeated pairs consequently has maximum degree one, and therefore has an independent set of size at least $\lceil |D|/2 \rceil \geq 4$. The four selected vertices are pairwise codegree one, so their trace is one of the six labelled K_4 clique partitions. Proposition 6 closes all six with maximum $16671/16796 < 1$.

5.5 Order 25

The degree excess above six is now

$$160 - 25 \cdot 6 = 10,$$

so the degree-six set D has size at least 15. For $x \in D$, (5) gives $\rho_x = 0$; every pair incident with D therefore has codegree exactly one.

Suppose first that an edge e contains at least four points of D . Choose four of them and a fifth vertex $z \in D \setminus e$, which is possible since $|D| \geq 15$. The trace of e on this selected set is a K_4 . Each of the four pairs joining z to that K_4 must be a separate K_2 trace, because combining two would reuse their internal K_4 pair. Thus the trace type is exactly $K_4 + 4K_2$.

Otherwise every edge contains at most three points of D , and hence covers at most one triple of D . There are at most 32 covered triples. If every five-set of D contained a covered triple, double-counting pairs (T, S) with $T \subseteq S$, $|T| = 3$, $|S| = 5$, would give

$$\binom{|D|}{5} \leq 32 \binom{|D|-3}{2}.$$

But

$$\frac{\binom{|D|}{5}}{\binom{|D|-3}{2}} = \frac{\binom{|D|}{3}}{10} \geq \frac{\binom{15}{3}}{10} = \frac{91}{2} > 32,$$

a contradiction. Therefore some five-set $S \subseteq D$ contains no covered triple. Since all its pairs have codegree one, every nonsingleton trace has size two and the trace type is exactly $10K_2$.

The favourable-profile computation contains one $10K_2$ profile, five labelled $K_4 + 4K_2$ profiles, and one K_5 regression profile. The selection above needs only the first two types; all are closed, and the largest value among the seven is $4188/4199 < 1$. This excludes $v = 25$ and completes the proof of Theorem 1.

6 A scoped optimization ceiling

It is natural to ask whether a richer search over locked positions, with no new combinatorial idea, would prove $m(5) \geq 34$. The following exhaustive calculation answers that narrower question negatively. Write $Q_{33}(l, p)$ for the evaluator in (12) applied to a trace profile l whose counts sum to 33. The result below is not an upper bound on $m(5)$ and does not assert that the displayed safe profiles are realisable.

For context, the same normalization with 33 edges leaves exactly the orders considered below. Pair coverage gives $\binom{v}{2} \leq 330$, hence $v \leq 26$, and (2) excludes $v \leq 18$. At $v = 19$, pair-incidence surplus gives at least $\lceil (330 - \binom{19}{2})/6 \rceil = 27$ unordered edge pairs meeting in at least two vertices. Thus $\gamma \leq 33 \cdot 32 - 2 \cdot 27 = 1002$. Substituting $5 \cdot 33 = 165$ for 160 in the right-hand side of (13), the corresponding zero-lock failure-probability bound is

$$\frac{228291}{230945} < 1.$$

At $v = 26$, pair coverage forces minimum degree seven, contradicting $26 \cdot 7 > 5 \cdot 33 = 165$. A 33-edge extension of the method must therefore handle precisely $20 \leq v \leq 25$.

Proposition 7 (ceiling of position optimization). *Consider the locked-permutation bound (12) using only selected degrees and their trace counts.*

- (i) For a hypothetical 33-edge core on 25 vertices, the safe five-lock profile with selected degrees (6, 6, 6, 6, 6) and trace partition $10K_2$ satisfies

$$\min_{\substack{p_1, \dots, p_5 \in [25] \\ \text{distinct}}} Q_{33}(l, p) = \frac{8649}{8398} > 1.$$

This minimum was obtained over all $25 \cdot 24 \cdot 23 \cdot 22 \cdot 21 = 6,375,600$ labelled placements; symmetry reduces the enumeration to $\binom{25}{5} = 53,130$ position sets. For comparison, the 32-edge certificate used in Section 5.5 gives $4188/4199 < 1$ for the corresponding profile.

- (ii) Each of the five labelled $K_4 + 4K_2$ profiles forced by the other branch of the selection in Section 5.5 has global minimum

$$\min_p Q_{33}(l, p) = \frac{325}{323} > 1.$$

One representative minimizing assignment is (12, 1, 13, 14, 15). The automorphism group has order 24, so 265,650 quotient representatives cover all 6,375,600 labelled placements for each labelled trace profile. At 32 edges the corresponding closing value is $37/38 < 1$.

- (iii) At each order $v = 20, 21, \dots, 25$, the safe three-lock universe for a 33-edge core contains a profile whose global minimum over all labelled three-lock placements is at least one.

The exact witnesses for the last assertion are listed in mask order (15):

v	degrees; counts	minimizing positions	global minimum
20	(8, 8, 8); (15, 4, 4, 2, 4, 2, 2, 0)	(1, 2, 12)	2287/2210
21	(7, 7, 7); (15, 5, 5, 1, 5, 1, 1, 0)	(1, 2, 3)	119365/116688
22	(7, 7, 7); (15, 5, 5, 1, 5, 1, 1, 0)	(1, 2, 3)	387055/369512
23	(7, 7, 7); (15, 5, 5, 1, 5, 1, 1, 0)	(1, 2, 3)	10671/9880
24	(6, 6, 6); (19, 4, 4, 0, 4, 0, 0, 2)	(1, 12, 14)	89966/88179
25	(6, 6, 6); (19, 4, 4, 0, 4, 0, 0, 2)	(1, 13, 15)	25888/24871

Consequently, both trace families forced by the 32-edge $v = 25$ selection fail when evaluated at 33 edges, and enlarging only the fixed position menus cannot turn that safe-profile argument into a proof of $m(5) \geq 34$.

The exhaustive counts and exact fractions in Proposition 7 are regenerated by the optimization audit in the artifact. The proposition does *not* exclude a proof of $m(5) \geq 34$ using stronger realizability constraints, a larger selected set, sharper counting of compatible witness edge pairs, or a different colouring argument. Rather, it identifies why further tuning of the current position menus is mathematically insufficient.

7 Reproducibility and independent verification

7.1 Artifact contents and exact outputs

The artifact contains three primary programs under **theory/**:

- `locked_greedy_32.py`, implementing (6)–(12);
- `close_v21_v22.py`, emitting one JSONL record for every initial or Ramsey-step profile at $v = 21, 22$; and

- `selection_certificates.py`, regenerating every $v = 23, 24, 25$ profile, clique partition, selected assignment and exact bound.

The $v = 21, 22$ artifact has 3,081 records. Its canonical internal SHA-256 and the SHA-256 of the JSONL file are recorded in `theory/certificates/v21_v22_manifest.json` and in the frozen release checksum file. The compact manifest `theory/SELECTION_MANIFEST.json` pins the six K_4 partitions, 32 K_5 partitions, every $v = 23$ profile and residual set, and every $v = 23, 24, 25$ selected certificate row. These latter finite sets are regenerated from compact source rather than archived as a second large row-by-row certificate.

The exact placement sweeps behind Proposition 7 and the $v = 21$ candidate-menu minimality statement, including the full labelled counts, automorphism quotients and exact minima, are documented in `optimization/OPTIMIZATION_AUDIT.md` and replayed by the accompanying optimization verifiers.

A second implementation under `solver_alt/` does not import the primary evaluator. It rederives the placement counts, enumerates three-lock profiles from different loop variables, and generates clique partitions as set partitions of $E(K_s)$ followed by a clique predicate. It replays all 3,081 $v = 21, 22$ rows and every $v = 23, 24, 25$ assertion, and compares the canonical row hashes. Hashes are integrity checks for the finite objects; they are not substitutes for the mathematical arguments in Sections 2, 3, and 5.

7.2 Commands and reference environment

From the artifact root, the lower-bound certificate is regenerated with

```
python3 theory/locked_greedy_32.py --self-test
python3 theory/close_v21_v22.py --output-dir theory/certificates
python3 theory/selection_certificates.py > theory/SELECTION_MANIFEST.json
python3 solver_alt/verify_v21_v22.py \
  --artifact-dir theory/certificates
python3 solver_alt/verify_lower33.py
```

The position-optimality and 33-edge ceiling calculations are regenerated from `optimization/` with

```
python3 -0 general_locked.py
python3 -0 compiled_locked.py
PYTHONPATH=. python3 -0 verify_v25_selection_ceiling.py \
  --output v25_selection_ceiling_verification.json
PYTHONPATH=. python3 -0 sweep_m32_v21_menu.py \
  --summary m32_v21_menu_summary.json \
  --records m32_v21_menu_records.jsonl
PYTHONPATH=. python3 -0 verify_m32_v21_base_subsets.py \
  --summary m32_v21_base_subsets_summary.json \
  --records m32_v21_six_base_profiles.jsonl
PYTHONPATH=. python3 -0 frontier_audit.py \
  --output optimization_results.json
```

No third-party Python package is required. A reference replay used CPython 3.13.1 on macOS/Darwin 25.5 (arm64), an 8-core Apple M3 and 16 GB RAM. The programs are single-process and exact-integer performance is sensitive to Python version and CPU scheduling. No minimum-memory claim is made; the complete pipeline was verified within 16 GB. The one-command driver

prints each command, verifies the checksum files, and reports the Python version and total wall time. Its pinned reports deliberately omit host-dependent paths and timings so that byte-for-byte comparisons remain portable.

7.3 Scope and limitations of the computer check

The finite proof has four explicit trust boundaries.

1. Python’s arbitrary-precision integers, `Fractions.Fraction` and `math.comb` are trusted; no proof-assistant kernel checks the arithmetic.
2. The primary and independent implementations share only the displayed mathematics and pinned finite outputs, but they run on the same Python runtime. Their algorithmic independence reduces coding risk without eliminating platform risk.
3. The profile universes are safe supersets. Unrealizable profiles can make the calculation harder but cannot create a false strict certificate for a realizable obstruction.
4. The computational propositions verify the finite inequalities; they do not replace the human-checkable selection arguments that connect a hypothetical hypergraph to those finite families.

Artifact availability. The code, exact certificates and verification reports supporting this article are available from the public GitHub repository `property-b-m5-lower33` (owner `AyushAg1002`) under release tag `v1.0.0`.

The corresponding archival release is identified by Zenodo DOI `10.5281/zenodo.22070117`. Source code is licensed under the MIT License; the manuscript source, proof notes, documentation, certificates, manifests and verification reports are licensed under CC BY 4.0. The DOI-bearing release contains the exact versions of all sources, certificate files, manifests, independent reports and optimization-audit materials cited in this paper.

8 Concluding remarks

Theorem 1 narrows the fifth Property-B interval to

$$33 \leq m(5) \leq 51.$$

The central methodological point is that a very small selected trace can carry enough information to certify all remaining vertex orders, provided that exact position-dependent probabilities are combined with degree and codegree selection. Proposition 7 also shows where the present certificate families stop: a proof of $m(5) \geq 34$ will require additional structural information or a stronger probability estimate, not merely a larger menu of locked placements for those families. The classical upper bound $m(5) \leq 51$ remains unchanged; we make no upper-bound claim.

Declarations

Use of generative AI and author responsibility. OpenAI Codex assisted with research and manuscript preparation; the author independently verified all proofs, computations, code, citations and wording and accepts full responsibility for the article.

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